Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

ANS:

C:\Users\User> ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : localdomain
IPv4 Address. . . . . : 192.168.1.15
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1

Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

ANS:

ping www.google.com

O/P

Ping statistics for 142.250.190.46:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 14ms, Maximum = 22ms, Average = 17ms

Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

ANS:

Tracing route to e4848.dscb.akamaized.net [23.51.10.12]
over a maximum of 30 hops:

1	<1 ms	<1 ms	<1 ms	192.168.1.1 (Home Router)
2	*	*	*	Request timed out.
3	4 ms	3 ms	3 ms	10.45.120.1 (ISP Gateway)
4	8 ms	7 ms	7 ms	172.16.25.14 (ISP Core Router)
...				
11	14 ms	13 ms	13 ms	23.51.10.12 (Flipkart CDN / Server Endpoint)

Trace complete.

Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

ANS:

Running nslookup www.spotify.com resolves to an IP address such as **35.186.224.24**.

- **What it verifies:** nslookup helps you verify that a domain name is correctly translating and resolving to its associated IP address via the Domain Name System (DNS).

A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Hint: Think about checking both network connectivity and DNS resolution.

ANS:

To diagnose why your friend cannot access Zomato's website, you can run the following sequence of three troubleshooting commands:

1. **`ipconfig`**
    - **What it checks:** Verifies the laptop's local network configuration. It ensures the computer has a valid local IPv4 address, subnet mask, and default gateway, confirming it is properly connected to the local Wi-Fi or router.
2. **`nslookup [www.zomato.com](https://www.zomato.com)`**
    - **What it checks:** Tests DNS (Domain Name System) resolution. It verifies whether your computer can successfully translate Zomato's web address into its corresponding IP address. If this fails, the browser won't know where to direct the traffic.
3. **`ping [www.zomato.com](https://www.zomato.com)`** (or **`tracert [www.zomato.com](https://www.zomato.com)`**)
    - **What it checks:** Tests active end-to-end network connectivity. It sends packets to Zomato's servers to see if they are reachable, measures response times (latency), and checks for packet loss along the route.